

Chapter 7 --- Modifying Database Structure

Once a table has been created, its structure is rarely final.

Applications grow, and new requirements appear --- a new column may be needed, an existing column's type may need to change, or an entire table may need to be renamed or emptied. The ALTER TABLE command is the DDL tool used to modify an existing table's structure without rebuilding it from scratch.

7.1 Adding, Modifying and Removing Columns Using ALTER TABLE

Starting table used for all examples in this chapter:

```
CREATE TABLE Students ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Branch VARCHAR(20));
```

Adding a Column --- ADD COLUMN

Adds a new column to an existing table.

```
ALTER TABLE Students ADD COLUMN Email VARCHAR(100);
```

You can also control where the new column appears using AFTER or FIRST:

```
ALTER TABLE Students ADD COLUMN Age INT AFTER StudentName;
```

You can add more than one column in a single statement:

```
ALTER TABLE Students ADD COLUMN City VARCHAR(30), ADD COLUMN Pincode VARCHAR(10);
```

Modifying a Column --- MODIFY COLUMN

Changes the data type or definition of an existing column, while keeping its name the same.

```
ALTER TABLE Students MODIFY COLUMN Email VARCHAR(150);
```

MODIFY can also add or remove constraints such as NOT NULL:

```
ALTER TABLE Students MODIFY COLUMN Branch VARCHAR(20) NOT NULL;
```

Renaming and Changing a Column Together --- CHANGE COLUMN

CHANGE lets you rename a column and redefine its type in the same statement (MODIFY cannot rename a column).

```
ALTER TABLE Students CHANGE COLUMN Email StudentEmail VARCHAR(150);
```

Removing a Column --- DROP COLUMN

Permanently removes a column and all the data stored in it.

```
ALTER TABLE Students DROP COLUMN Pincode;
```

DROP COLUMN cannot be undone --- the data in that column is permanently lost once the statement runs.

Quick Reference --- ALTER TABLE Column Operations

Goal Syntax Can Rename?

Add a column ADD COLUMN col_name ---
datatype

Change a column's MODIFY COLUMN col_name No
type/constraints new_datatype

Rename + redefine a CHANGE COLUMN old_name Yes
column new_name datatype

Remove a column DROP COLUMN col_name ---

7.2 Renaming Columns and Tables

Renaming a Table

Two equivalent ways to rename an entire table:

```
RENAME TABLE Students TO Learners;
```

or

```
ALTER TABLE Students RENAME TO Learners;
```

Renaming a Column

MySQL 8.0 and later supports a direct RENAME COLUMN clause:

```
ALTER TABLE Students RENAME COLUMN StudentName TO FullName;
```

On older MySQL versions, use CHANGE COLUMN instead (you must repeat the data type):

```
ALTER TABLE Students CHANGE COLUMN StudentName FullName VARCHAR(50);
```

Renaming a table or column does not affect the data inside it --- only its name changes. However, any queries, views, or application code that reference the old name must be updated separately.

7.3 Understanding TRUNCATE and DROP

TRUNCATE, DROP, and DELETE are sometimes confused because they can all appear to "remove data" --- but they operate very differently.

TRUNCATE TABLE

Removes all rows from a table instantly, but keeps the table's structure (columns, data types, constraints) intact so it can be reused immediately.

```
TRUNCATE TABLE Students;
```

- Deletes all rows --- cannot be used with a WHERE clause.
- Resets any AUTO_INCREMENT counter back to its starting value.
- Is a DDL operation --- much faster than DELETE for clearing an entire table, and generally cannot be rolled back.
- The table itself still exists and can be used right away.

DROP TABLE

Permanently deletes the entire table --- its structure, its data, its indexes, and its constraints --- from the database.

```
DROP TABLE Students;
```

- The table no longer exists after this command runs.
- Any indexes, constraints, and permissions tied to the table are removed as well.
- Cannot be undone (without a backup).

DROP can also remove an entire database:

```
DROP DATABASE sql_course;
```

DELETE (for comparison)

Unlike TRUNCATE and DROP, DELETE is a DML command that removes rows one at a time and can be filtered with WHERE, making it reversible inside a transaction.

```
DELETE FROM Students WHERE Branch = 'ECE';
```

TRUNCATE vs DROP vs DELETE --- Comparison

Feature TRUNCATE DROP DELETE

Command type DDL DDL DML

What it removes All rows only Entire table Selected rows (or
(structure + all, if no WHERE)
data)

Table structure Still exists, No longer exists Still exists
after empty

Supports WHERE No No Yes
clause

Speed Very fast Very fast Slower
(row-by-row)

AUTO_INCREMENT Reset to start Removed with the Not reset
counter table

Can be rolled Generally no Generally no Yes, within a
back transaction

■ How to Remember It TRUNCATE empties the room but keeps the house standing. DROP demolishes the house completely. DELETE removes chosen items from the room, one at a time, and lets you put them back if you change your mind before saving.

7.4 Interview Questions --- Chapter 7

Q1. What is the difference between MODIFY and CHANGE in ALTER TABLE?

Ideal answer: MODIFY changes a column's data type or constraints but

keeps its name the same. CHANGE can rename the column and redefine its type in the same statement --- MODIFY alone cannot rename a column.

Q2. What is the difference between TRUNCATE and DELETE?

Ideal answer: TRUNCATE is a DDL command that removes all rows at once, resets AUTO_INCREMENT, and cannot use a WHERE clause. DELETE is a DML command that removes rows one at a time, can be filtered with WHERE, and can be rolled back within a transaction.

Q3. What is the difference between TRUNCATE and DROP?

Ideal answer: TRUNCATE empties a table but keeps its structure so it can be reused immediately. DROP removes the table entirely --- its structure, data, indexes, and constraints --- and the table no longer exists afterward.

Q4. How do you add a new column to an existing table?

Ideal answer: Using ALTER TABLE with ADD COLUMN, e.g. ALTER TABLE Students ADD COLUMN Email VARCHAR(100);

Interview Questions — Chapter 7

Q1. MODIFY vs CHANGE in ALTER TABLE?

Ideal answer: MODIFY changes a column's definition while keeping its name. CHANGE can rename the column and redefine its type.

Q2. TRUNCATE vs DROP?

Ideal answer: TRUNCATE removes all rows but keeps the table structure; DROP removes the table itself, including its structure and data.

Q3. Can TRUNCATE use WHERE?

Ideal answer: No. TRUNCATE removes all rows.

Q4. What does ALTER TABLE do?

Ideal answer: It changes an existing table's structure, such as adding, modifying, renaming, or removing columns.

Q5. Does renaming a column automatically update application code?

Ideal answer: No. Queries, views, procedures, or application code that reference the old name may need separate updates.